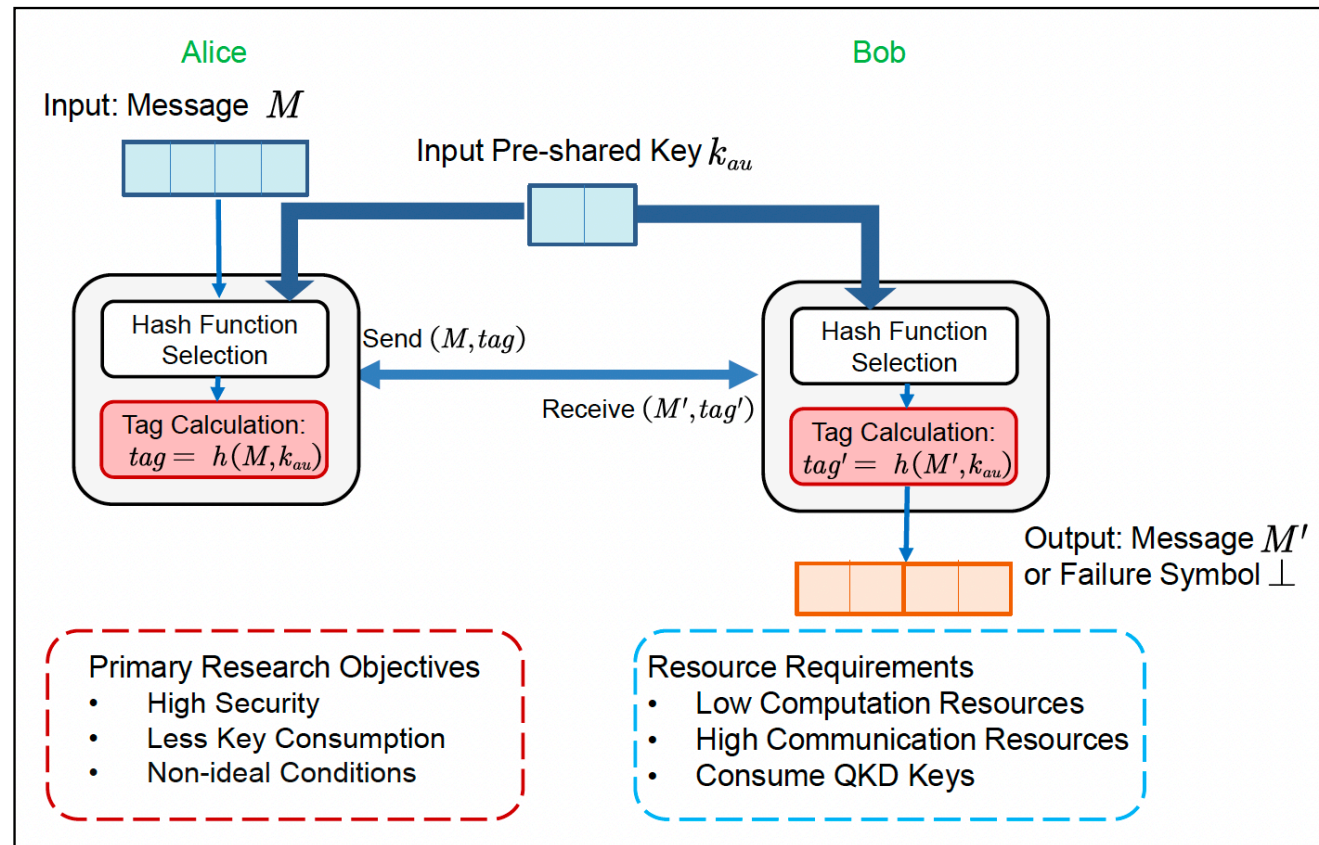




Authentication

- Authentication ensures the integrity and security of the communication between Alice and Bob.
- Classical Channel is authenticated using some preshared keys
- A part of final QKD keys (from previous rounds) is consumed for authentication

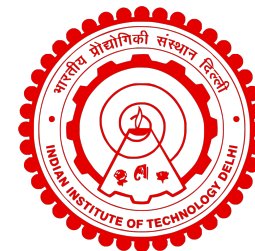


Outline



1. Introduction to Cryptography
2. Quantum Key Distribution (QKD)
3. Discrete Variable QKD
4. Continuous Variable (CV) Quantum Information
5. Gaussian Quantum States
6. Continuous Variable Quantum Key Distribution (CVQKD)
7. Secret Key Rate Analysis

Quantization of EM Field



1. Electric Field: $\mathbf{E}(r, t) = \sum_{k,s} \left(\frac{h\omega_k}{4\pi\epsilon_0} \right)^{1/2} \mathbf{e}_k^{(s)} \left[\alpha_{k,s} e^{i(kr - \omega_k t)} + \alpha_{k,s}^* e^{-i(kr - \omega_k t)} \right]$
2. Promote Fourier coefficients $\alpha_{k,s}$ into operators $\hat{a}_{k,s}$ s.t. $\xrightarrow{[\hat{a}, \hat{b}] = \hat{a}\hat{b} - \hat{b}\hat{a}}$
3. Annihilation ($\hat{a}$) and creation operators ($\hat{a}^\dagger$)
4. QM representation for single mode: $\hat{\mathbf{E}}(r, t) = E_0 \mathbf{e} [\hat{X}_1 \cos(kr - \omega t) + \hat{X}_2 \sin(kr - \omega t)]$
5. Quadrature Components: $\hat{X}_1 = (\hat{a} + \hat{a}^\dagger)$, $\hat{X}_2 = i(\hat{a}^\dagger - \hat{a})$
6. Annihilation operator $\hat{a} = \hat{X}_1 + i\hat{X}_2$ analogous to complex baseband signal
7. For single mode, quantized EM field is analogous to quantum harmonic oscillator
8. Electric and magnetic field plays the roles of position and momentum

$$\begin{aligned} [\hat{a}_{k,s}, \hat{a}_{k',s'}^\dagger] &= \delta_{kk'} \delta_{ss'} \\ [\hat{a}_{k,s}, \hat{a}_{k',s'}] &= 0 \\ [\hat{a}_{k,s}^\dagger, \hat{a}_{k',s'}^\dagger] &= 0 \end{aligned}$$



Bosonic System-I

1. A quantum system is called continuous-variable system when it has an infinite-dimensional Hilbert space described by observables with continuous eigenspectra.
2. A continuous-variable system is represented by N bosonic modes, corresponding to N quantized radiation modes of the electromagnetic field.
3. Tensor product Hilbert space: $\mathcal{H}^{\otimes N} = \bigotimes_{k=1}^N \mathcal{H}_k$ and corresponding N pairs of annihilation and creation operators: $\left\{ \hat{a}_k, \hat{a}_k^\dagger \right\}_{k=1}^N$
4. In vector operator: $\hat{\mathbf{b}} = \left(\hat{a}_1, \hat{a}_1^\dagger, \dots, \hat{a}_N, \hat{a}_N^\dagger \right)^T$ with commutator relation: $[\hat{b}_i, \hat{b}_j] = \Omega_{ij}$ where
$$\Omega := \bigoplus_{k=1}^N \omega = \begin{bmatrix} \omega & & \\ & \ddots & \\ & & \omega \end{bmatrix}, \quad \omega := \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix},$$
5. Fock states $\{ |n\rangle \}_{n=0}^\infty$: Eigenstates of number operator $\hat{N} = \hat{a}^\dagger \hat{a}$ s.t. $\hat{N}|n\rangle = n|n\rangle$
6. Fock states are orthogonal, and form a basis for the Hilbert space: $\langle n | m \rangle = \delta_{nm}$



Bosonic System-II

1. Quadrature field operators: $\left\{ \hat{q}_k, \hat{p}_k^\dagger \right\}_{k=1}^N$ where $q_k = a_k + a_k^\dagger$, $p_k = i(a_k^\dagger - a_k)$ and $\hat{\mathbf{x}} := (\hat{q}_1, \hat{p}_1, \dots, \hat{q}_N, \hat{p}_N)^T$. Then the commutator relation is: $[\hat{x}_i, \hat{x}_j] = 2i\Omega_{ij}$
2. The quadrature operator $\hat{q}(\hat{p})$ (which is Hermitian) is an observable with continuous eigenspectra, i.e., $\hat{q}|q\rangle = q|q\rangle$ ($\hat{p}|p\rangle = p|p\rangle$), with orthogonal eigenvectors or eigenstates $|q\rangle$ ($|p\rangle$) having continuous eigenvalues $q \in \mathbb{R}$ ($p \in \mathbb{R}$).
3. **Phase Space representation of CV quantum states:**
4. A quantum state $\hat{\rho}$ of a N -mode CV system can also be described in terms of a characteristic function $\chi_c(\xi) = \text{Tr}(\hat{\rho}\hat{D}(\xi))$, where $\hat{D}(\xi) = \exp(i\hat{x}^T\Omega\xi)$ is the Weyl operator.
5. Wigner Function (quasi-probability distribution): $W(\mathbf{x}) = \int_{\mathbb{R}^{2N}} \frac{d^{2N}\xi}{(2\pi)^{2N}} \exp(-i\mathbf{x}^T\Omega\xi) \chi(\xi)$, : Fourier Transform of characteristic function $\chi_c(\xi)$
6. The CV quantum states can be visualized using their Wigner function in a phase-space representation, where the axes are defined by a pair of conjugate quadrature variables q and p .

Gaussian Quantum States - I

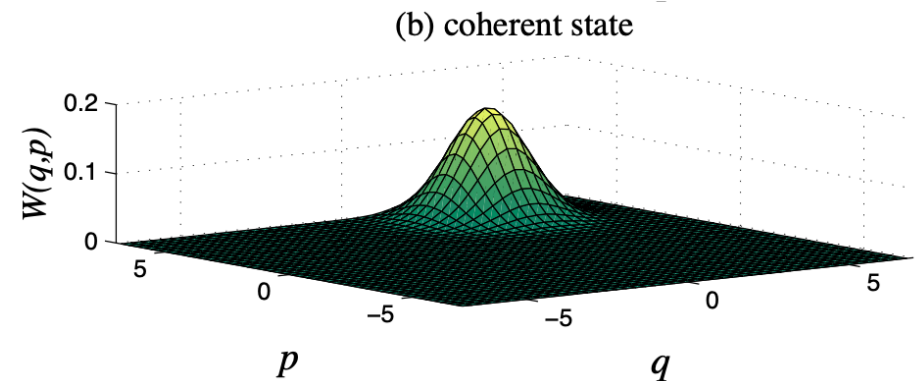
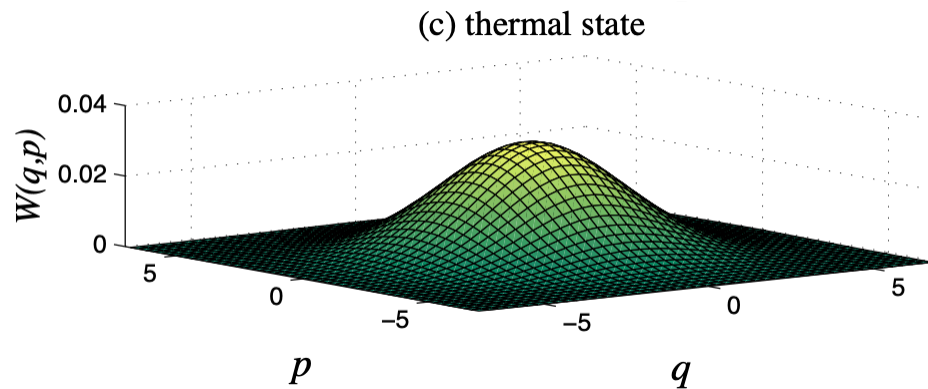
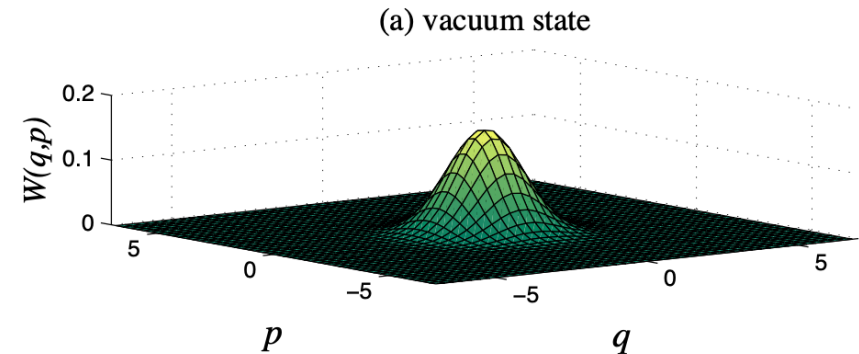


1. The Wigner function is characterised by the statistical moments:
2. Mean Displacement Vector: $\bar{\mathbf{x}} := \langle \hat{\mathbf{x}} \rangle = \text{Tr}(\hat{\mathbf{x}}\hat{\rho})$,
3. Covariance Matrix ($\mathbf{V}$): $\mathbf{V}_{i,j} = \frac{1}{2} \langle \{ \Delta \hat{x}_i, \Delta \hat{x}_j \} \rangle$ where $\Delta \hat{x}_i := \hat{x}_i - \langle \hat{x}_i \rangle$ and $\{a, b\} = ab + ba$ is anti-commutator. Diagonal entries are variances: $\mathbf{V}_{i,i} = V(\hat{X}_i) = \langle \Delta \hat{X}_i^2 \rangle$
4. Uncertainty principle: $\mathbf{V} + i\mathbf{\Omega} \geq 0$
5. Heisenberg's uncertainty principle for diagonal entries: $V(\hat{q}_k) V(\hat{p}_k) \geq 1$.
6. Gaussian states: Quantum states completely characterised by first two moments : $\hat{\rho} = \hat{\rho}(\bar{\mathbf{x}}, \mathbf{V})$
7. Wigner Function is Gaussian:
$$W(\mathbf{x}) = \frac{\exp \left[-(1/2)(\mathbf{x} - \bar{\mathbf{x}})^T \mathbf{V}^{-1} (\mathbf{x} - \bar{\mathbf{x}}) \right]}{(2\pi)^N \sqrt{\det \mathbf{V}}}$$
8. Gaussian Channels: Transform Gaussian states into Gaussian states: $\bar{\mathbf{x}} \rightarrow \mathbf{S}\bar{\mathbf{x}} + \mathbf{d}, \quad \mathbf{V} \rightarrow \mathbf{S}\mathbf{V}\mathbf{S}^T$.

Gaussian Quantum States - II



1. **Vacuum State:** Zero mean and identity covariance matrix (quantum shot noise)
2. **Coherent State:** Eigenstates of annihilation operator $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$, $\alpha \in \mathbb{C}$. Mean: (q, p) and identity covariance matrix where $\alpha = (q + ip)/2$
3. **Thermal State:** Mixture of coherent states. Zero mean and covariance matrix: $V_{\text{th}} = (2\bar{n} + 1)I$ with $\bar{n} > 0$ is the average number of photons



Gaussian Quantum States - II

1. **Single Mode Squeezed Vacuum State:** Zero mean and diagonal covariance matrix :
 $V = \text{diag}[\exp(-2r_s), \exp(2r_s)]$ where $r_s \in [0, \infty)$ is the single-mode squeezing parameter.

1. For $r_s > 0$: $V(\hat{q}) < 1$ and $V(\hat{p}) > 1$ s.t. Heisenberg's uncertainty principle is satisfied

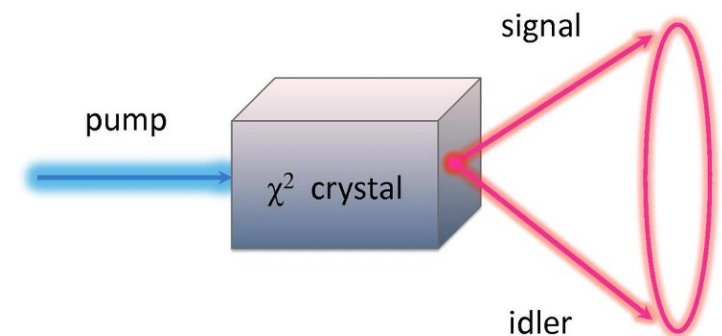
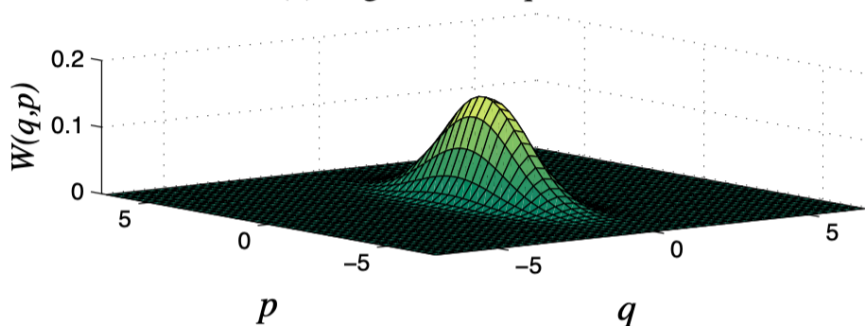
2. **Two mode Squeezed Vacuum State (TMSV): Entangled State**

1. $|\psi\rangle_{TMSV} = \sqrt{1 - \lambda^2} \sum_{n=0}^{\infty} (-\lambda)^n |n\rangle_1 |n\rangle_2$, where $\lambda = \tanh(\xi)$

2. Covariance matrix: $V_{TMSV} = \begin{pmatrix} \nu \mathbf{I}_2 & \sqrt{\nu^2 - 1} \mathbf{Z}_2 \\ \sqrt{\nu^2 - 1} \mathbf{Z}_2 & \nu \mathbf{I}_2 \end{pmatrix}$, where $\mathbf{Z}_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\nu = \cosh(2\xi)$

3. The quadratures of the two modes are correlated

(d) single-mode squeezed state



Von Neumann Entropy of Gaussian States



1. Von-Neumann (VN) Entropy: $S(\rho) = -\text{Tr}(\rho \log \rho) = -\sum_i \lambda_i \log \lambda_i$

2. For Gaussian states, VN entropy depends on covariance matrix

3. Symplectic Eigenvalues: For an N mode covariance matrix $\mathbf{V}$, there exists a symplectic matrix $\mathbf{S}$ s.t. $\mathbf{V} = \mathbf{S}\mathbf{V}^\oplus\mathbf{S}^T$, $\mathbf{V}^\oplus := \bigoplus_{k=1}^N \nu_k \mathbf{I}_2$

4. Symplectic matrix $\mathbf{S}$: $\mathbf{S}\mathbf{\Omega}\mathbf{S}^T = \mathbf{\Omega}$ where $\mathbf{\Omega} := \bigoplus_{k=1}^N \omega = \begin{pmatrix} \omega & & \\ & \ddots & \\ & & \omega \end{pmatrix}$, $\omega := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

5. Symplectic spectrum equals standard eigenspectrum of $|i\mathbf{\Omega}\mathbf{V}|$

6. VN Entropy:

$$S(\hat{\rho}) = \sum_{k=1}^N h(\nu_k), \text{ where } h(x) := \left(\frac{x+1}{2}\right) \log \left(\frac{x+1}{2}\right) - \left(\frac{x-1}{2}\right) \log \left(\frac{x-1}{2}\right).$$